# When Does Relational Structure Help Healthcare Fraud Detection? A Controlled Benchmark of Heterogeneous Graph Neural Networks and Graph-Augmented Tabular Models

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**Abstract**

**Purpose:** Graph neural networks (GNNs) are proposed for healthcare insurance fraud because collusive provider rings leave traces in claim networks, but when they beat tabular models is unclear. **Methods:** We built a synthetic benchmark that independently varies the feature-level signal and the doctor–patient density of fraud rings, tests on rings of unseen doctors and patients, and analyses audit periods separately to avoid leakage. We compared 14 equally tuned methods, including gradient boosting with and without engineered graph features and five heterogeneous GNNs, over 54 worlds in nine regimes, evaluated routing from gradient boosting to a GNN, and checked the findings against public provider-labelled Medicare-style data (5,410 providers). **Results:** Without a feature-level signal, claim-only models were at chance, while per-entity aggregates detected most rings (average precision 0.89–0.93); structure added up to 0.10, only for moderately or densely interlinked rings. Pooled over worlds, no GNN outperformed gradient boosting with engineered graph features (GraphSAGE −0.0001, HGT −0.011), and GNNs were 0.08–0.15 worse on isolated fraud. Routing high-risk claims to a GNN gained nothing; fusing both models' rankings gained at most 0.006. Test rings reusing training-ring members inflated performance by up to 0.60. On real data, gradient boosting on provider aggregates was best (0.704);

graph information did not help. **Conclusion:** Relational modelling pays off only for dense collusion and then mainly through structure that simple engineered features capture. Graph models should be evaluated against graph-augmented tabular baselines on unseen entities.

**Keywords:** Healthcare fraud detection, Provider rings, Heterogeneous graph neural networks, Gradient boosting, Synthetic benchmark, Data leakage

# 1 Introduction

Health insurance fraud causes losses estimated in the tens of billions of dollars each year in the United States alone [1]. It takes two broad forms. In *isolated* fraud a single provider inflates or fabricates individual claims, so the evidence lies in the claim itself: amounts, procedure counts or coding that are unusual for the provider's specialty. In *collusive* fraud a group of providers and recruited patients file claims together. Each claim can look ordinary; the evidence lies in who bills whom—patients who see many members of the same small group of providers, and providers who share an unusually large fraction of their patients.

The second form motivates graph neural networks (GNNs) over the patient–provider–facility network, in particular heterogeneous architectures such as RGCN [2], HAN [3], HGT [4] and Simple-HGN [5]. Whether GNNs are needed, however, is an empirical question. Gradient-boosted trees remain strong on tabular data [6], relational signals can be engineered as features, and benchmark studies outside healthcare report that tree ensembles with neighbourhood features often match or beat GNNs on graph anomaly detection [7, 8].

Answering the question for healthcare claims is difficult for two reasons. First, public claims data with labelled collusion are not available, and real labels conflate fraud typologies. Second, evaluation is fragile: identifiers that encode the label, statistics computed on the full dataset, and providers that appear in both training and test data inflate apparent performance [9, 10]. Fraud rings make the last problem acute, because a model can score a test claim correctly by recognising a provider it saw labelled during training instead of recognising coordination.

We therefore study the question in a controlled setting and check it on real data. Our contributions are:

1. **A regime-controllable benchmark.** A synthetic claims generator in which the feature-level signal of ring claims ($\delta$) and the density of the ring's doctor–patient block ($s$) are varied independently, with an *emergent* protocol in which every test ring is formed by doctors and patients never seen in a training ring, and an audit-period design that keeps features and graphs free of future information (Section 3). Code, generator and tests are released.
2. **A controlled comparison** of 14 methods—tabular models with and without engineered graph features, a multilayer perceptron (MLP) control, five heterogeneous GNNs, two hybrids and an unsupervised overlap score—with an equal tuning budget, nine regimes, six independently generated worlds per regime and paired

statistics, together with an evaluation of two-stage routing from gradient boosting to a GNN (Sections 4 and 6).
3. **Mechanism and memorisation analyses**: degree-preserving rewiring, single-factor ablations of HGT (depth, attention heads, relations, node inputs), an analysis of which inputs the trained models rely on, and a comparison of new versus recurring rings that quantifies how much apparent ring detection comes from recognising known members (Section 7). A re-analysis of a small fixed benchmark shows how such data can suggest the opposite conclusion (Section 9.3).
4. **A real-data check** on a public, provider-labelled Medicare-style claims dataset with a provider-disjoint protocol (Section 5). Its labels identify providers, not rings, so it tests whether graph information helps provider-level detection, not whether rings are detected.

Our main findings are as follows. Claim features alone cannot reveal rings whose claims look legitimate, but per-entity aggregates capture most of the evidence, because ring members are unusually active. Coordination structure adds measurable value only when the ring's doctor–patient block is dense. Where it does, gradient boosting with a few engineered shared-patient features is as good as or better than every GNN we tested; GNNs that also receive those features gain a small but consistent amount (+0.008 in average precision, AP, pooled over all worlds). GNNs are clearly worse on isolated fraud, because message passing dilutes claim-level anomalies. HGT's ring detection is structural: it collapses without degree encodings and survives with degree encodings as its only inputs. Routing high-risk claims from gradient boosting to a GNN gained nothing; fusing the two models' rankings gained at most 0.006 AP, as an ensemble rather than as a router. Evaluations that let test rings share members with training rings overstate ring detection, by up to 0.60 AP for a model that only sees entity attributes. On the real data, where labels show no relational assortativity, no graph-based method improved on gradient boosting over provider aggregates.

## 2 Related Work

***Tabular healthcare fraud detection.***

Most published healthcare-fraud detectors are supervised classifiers over claim- or provider-level features. On Medicare data labelled with the federal exclusion list (LEIE), Bauder and Khoshgoftaar [11] compared classical learners, Herland et al. [12] combined several Medicare data sources, and Johnson and Khoshgoftaar [13] studied neural networks under extreme class imbalance. Earlier work on Medicaid data used multidimensional data models and outlier analysis [14]. These models score each claim or provider from its own attributes and pre-computed aggregates; relational information enters, if at all, through hand-engineered features.

***Graph-based fraud and anomaly detection.***

Surveys of graph-based anomaly detection [15, 16] document how coordination leaves structural traces such as dense blocks, lockstep behaviour and unusual egonets. Unsupervised detectors exploit these traces directly, e.g. OddBall [17], CopyCatch [18] and FRAUDAR [19]. Graph neural networks (GNNs) have been applied to review

fraud [20], underground-forum key players [21], and camouflaged or class-imbalanced fraudsters [22, 23]; spectral analyses argue that anomalies shift graph spectra in ways that standard message passing can blur [24].

***Heterogeneous GNNs.***

Relational GCN [2] uses relation-specific weights; HAN [3] aggregates over hand-chosen meta-paths with node- and semantic-level attention; HGT [4] parameterises attention by node and edge type without meta-paths; and Simple-HGN [5] showed that a GAT with edge-type embeddings, residual attention and output normalisation is competitive with more elaborate heterogeneous architectures when baselines are tuned fairly.

***When do graph models beat tabular models?***

Evidence on this question is mixed. On the Elliptic Bitcoin dataset a random forest on node features outperformed graph convolutional networks under a temporal split [7]. GADBench [8] found that tree ensembles with simple neighbourhood aggregation are among the strongest supervised graph anomaly detectors across ten datasets, and tree ensembles remain hard to beat on tabular data generally [6]. Synthetic benchmarks with injected typologies make the question testable: Altman et al. [25] generate money-laundering patterns with known ground truth, and Jesus et al. [26] release controlled tabular fraud suites. To our knowledge no comparable controlled study exists for heterogeneous patient–provider–facility claim graphs, where the relevant typology is the collusive provider ring.

***Evaluation pitfalls.***

Leakage through features that encode the target, through preprocessing fitted on the full dataset, or through entities shared between training and test data inflates reported performance [9, 10]. GNN benchmarks are additionally sensitive to split choice and tuning effort [27]. These risks are acute for fraud rings, whose members recur across claims: a model can score a test claim correctly by recognising a provider it saw labelled in training rather than by recognising coordination.

# 3 Benchmark Design

## 3.1 Synthetic claims world

We simulate 6,000 patients, 300 doctors and 40 hospitals in five regions over four audit periods of six months (Fig. 1; Table A1). Each doctor has one of eight specialties, which sets the distribution of claim amounts, inpatient probability and procedure counts. Patients have heterogeneous utilisation (gamma-distributed claim rates with mean three claims per two years) and one to three regular doctors in their region; 85% of legitimate claims go to a regular doctor and the rest to a popularity-weighted random doctor. Claim amounts are log-normal given the specialty, procedure count and inpatient status.

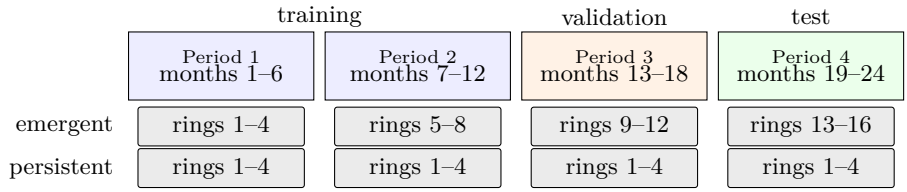



**Fig. 1** Audit-period protocol. Each six-month period is analysed separately: features and graphs use only that period's claims. Under the emergent protocol every period has new rings formed by disjoint doctors and patients, so test rings are unseen; under the persistent protocol the same rings recur and can be recognised by their members.

***Isolated fraud.***

Three per cent of claims are upcoded claims filed through a patient's own regular doctor. Their log-amount, procedure count, inpatient probability and procedure-code level are shifted by two standard deviations; their structure is that of an ordinary claim.

***Provider rings.***

A ring consists of four colluding doctors and twelve recruited patients, drawn preferentially from high utilisers. During its active period the ring files 16 claims per month between linked doctor–patient pairs, clustered around monthly burst days. Two parameters are varied independently:

- the *feature shift* $\delta \in \{0, 1, 2\}$ applied to ring claims, in standard deviations of log-amount (with matching shifts in procedures, inpatient probability and coding level). At $\delta = 0$ a ring claim is drawn from exactly the distribution of a legitimate claim of the same doctor's specialty;
- the *structural density* $s \in \{0.1, 0.5, 0.9\}$, the probability that a ring patient is linked to each ring doctor. Ring claim volume is fixed, so $s$ changes only how the claims are spread over the doctor–patient block (realised block densities $\approx 0.37$, 0.62 and 0.79).

Ring patients keep a single regular doctor, so the number of distinct doctors they see is governed by $s$. Their elevated claim volume, a degree signal visible to every model, is the same in all regimes.

***Protocols.***

Under the *emergent* protocol (default), four new rings start in each period and rings in different periods are formed by disjoint doctors and patients, so every test ring is unseen during training. Under the *persistent* protocol the same four rings are active in all periods, which reproduces the setting of most prior synthetic evaluations and lets a model succeed by recognising known members. Both protocols have identical claim and ring-claim volumes per period. Ground truth (fraud type and ring identifier) is stored separately from the claim records and never enters feature construction.

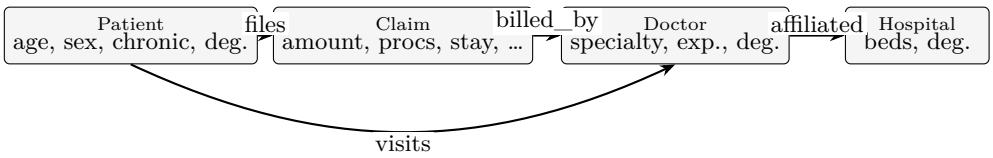



**Fig. 2** Heterogeneous claim graph of one audit period. Every relation also has a reverse relation (not drawn), so every node type receives messages and doctor context reaches claim nodes. Entity nodes carry static attributes and log(1 + degree) per incident relation; claim nodes carry only claim-intrinsic attributes.

### 3.2 Evaluation protocol

Periods 1–2 are used for training, period 3 for validation (early stopping, threshold selection) and period 4 for testing. Each period is analysed on its own, as in a retrospective audit: a claim's aggregate and graph features, and the graph a GNN operates on, are built only from claims of the same period. This keeps every split's features on the same time scale and prevents information from later periods reaching earlier ones; it is verified by automated tests (Section 10).

## 4 Methods

### 4.1 Tabular features

- **C** (claim): log-amount, procedures, length of stay, inpatient flag, coding level, month (cyclical), patient age, sex and chronic-condition count, doctor experience and specialty, hospital size.
- **A** (aggregates within the period, no labels): the patient's claim count, distinct doctors and mean/maximum log-amount; the doctor's claim count, distinct patients and mean log-amount; hospital claim count; patient–doctor pair count; and the claim's log-amount relative to the patient, doctor and specialty means.
- **G** (graph features on the period's patient–doctor bipartite graph, no labels): the doctor's maximum shared-patient Jaccard similarity with another doctor and number of doctors sharing at least two patients, the mean degree of the doctor's patients, the patient's mean doctor overlap and share of doctor pairs that co-treat other patients, and the triadic closure of the claim (how many of the patient's other doctors share patients with the claim's doctor), raw and degree-normalised.

### 4.2 Heterogeneous graph

Each period's graph has patient, doctor, hospital and claim nodes and the relations *files* (patient→claim), *billed_by* (claim→doctor), *affiliated* (doctor→hospital) and *visits* (patient→doctor), each with a reverse relation so that every node type receives messages. Claim nodes carry claim-intrinsic attributes; entity nodes carry their static attributes and log(1 + degree) for each incident relation, a label-free structural encoding included because attention- and mean-based aggregators cannot otherwise count neighbours [28]. No engineered aggregates or labels are used as node features (Fig. 2). Standardisation statistics are fitted on training data.

## 4.3 Models

- **Tabular**: logistic regression (LR) and random forest (RF) on C+A; histogram gradient boosting (GB) [29] on C, C+A and C+A+G.
- **MLP-C**: a multilayer perceptron on C, the no-graph control for the GNNs.
- **GNNs**: heterogeneous GraphSAGE [30] (one convolution per relation, mean or sum aggregation), RGCN [2], HAN [3] (claim nodes; meta-paths claim–patient–claim and claim–doctor–claim, with patient, doctor and hospital attributes appended to claim inputs), Simple-HGN [5] and HGT [4]. All share the same per-type input projection, residual connections with layer normalisation, and two-layer classification head.
- **Hybrids**: SAGE+AG and HGT+AG, whose claim nodes also receive the A and G features, testing whether message passing adds information beyond the engineered features. Hybrids reuse the tuned configuration of their base architecture and receive no separate tuning budget.
- **Overlap score**: the degree-normalised triadic closure of the claim, used directly as an unsupervised score.

Tabular models use scikit-learn [31] and graph models PyTorch Geometric [32]. Neural models are trained full-batch with AdamW, class-weighted binary cross-entropy and early stopping on validation average precision (patience 20, at most 200 epochs).

***Hyperparameters.***

Every method receives the same budget of up to ten configurations (the first being a default), selected by mean validation average precision on two development worlds (generator seeds 901 and 902, $\delta = 0.5$, $s = 0.5$) that are never used for evaluation (Appendix A, Table A2). The development regime ($\delta = 0.5$) is deliberately not one of the evaluation regimes.

## 4.4 Two-stage routing

A common deployment proposal is to score every claim with a cheap tabular model and pass only high-risk claims to a GNN. We evaluate this with GB-CAG as the first stage and HGT or SAGE as the second. A claim is escalated if its GB-CAG score is at least the $(1 - e)$ quantile of the GB-CAG scores in the validation period. The escalation fraction $e \in \{0.05, 0.1, 0.2, 0.3, 0.5, 1\}$ is chosen by validation AP, so test claims never influence it. Escalated claims rank above all other claims and are ordered either by the GNN score alone (*re-rank*: GB-CAG acts as a filter) or by the mean of the two models' validation percentiles (*fused*); non-escalated claims keep their GB-CAG order. With $e = 1$, re-ranking reduces to the GNN alone and fusion to a rank-average ensemble of the two models, so both single models and the ensemble are among the candidates. A claim's routed score depends only on its own two scores and on validation statistics. The stages are the models of the main experiment, retrained with the same seeds; their test predictions reproduce the stored ones exactly for the GNNs and to within $3 \times 10^{-8}$ for GB-CAG (float32 storage).

### 4.5 Metrics and statistics

The primary metric is average precision (AP, the area under the precision–recall curve), appropriate under class imbalance [33]. *Ring AP* scores ring claims against legitimate claims (isolated fraud excluded); *isolated AP* is defined analogously. We also report F1 at the threshold that maximises validation F1, ROC-AUC and recall when the top 5% of claims are investigated. For routing we add precision among the top 5% of claims (P@5%) and, as a ring-level measure, the share of test rings with at least one claim in that budget. Each of the nine regimes is replicated with six independently generated worlds (generator seeds 1–6), 54 worlds in total. Methods are compared within a regime with paired $t$-tests across worlds; $p$-values are Holm-adjusted within each pre-specified family of comparisons [34], and Wilcoxon signed-rank tests are reported as a robustness check [35]. Intervals are 95% $t$-intervals over worlds.

### 4.6 Use of AI tools

An AI coding assistant (Claude, Anthropic) was used to help write and review the analysis code, run and document the experiments, and draft and edit parts of the text. All reported numbers are produced by the released code (Section 10). The authors reviewed all code, analyses and text and take full responsibility for the content.

## 5 Real-World Data

### 5.1 Data

We use the public "Healthcare Provider Fraud Detection Analysis" dataset [36] (CC0 public domain), which links Medicare-style inpatient and outpatient claims to beneficiaries and labels each provider as potentially fraudulent or not. Only the labelled training files are used: 558,211 claims (40,474 inpatient, 517,737 outpatient) dated between November 2008 and December 2009, filed by 5,410 providers (506, or 9.4%, labelled fraudulent) for 138,556 beneficiaries, with 100,737 distinct attending, operating or other physicians. The data documentation does not describe how the labels were obtained, so we treat them as given and discuss this limitation in Section 9.6. Because labels exist only per provider, the prediction unit is the provider; claim-level ring detection cannot be evaluated on these data.

### 5.2 Features and graph

**P** (23 provider aggregates): claim volume and inpatient share, beneficiaries and claims per beneficiary, distinct attending and operating physicians, reimbursement totals and log-amount statistics, deductibles, claim duration and length of stay, diagnosis and procedure counts, share of claims with an operating physician, and the age, chronic-condition burden, mortality, renal-disease share, annual reimbursements and geographic spread of the provider's beneficiaries. **G** (8 label-free graph features): for both shared beneficiaries and shared physicians, the number of co-providers, the maximum Jaccard overlap with another provider, the share of the provider's beneficiaries

(physicians) seen by other providers, and their mean number of providers. **L** (4 label-neighbourhood features): the share of co-providers that are known fraudulent *training* providers, and an indicator of having any labelled co-provider, through beneficiaries and through physicians. L uses only training labels and never a provider's own label.

The heterogeneous graph has provider, beneficiary and physician nodes, with relations *serves* (provider–beneficiary, 363,300 edges) and *employs* (provider–physician) plus reverse relations. Physicians working for a single provider carry no cross-provider information and are summarised in P rather than kept as nodes, leaving 7,685 shared physicians (16,287 edges). Provider nodes carry P (or P+G for the hybrids), beneficiary nodes 22 demographic and clinical attributes, and physician nodes their claim and provider counts. HAN uses the provider-only meta-paths provider–beneficiary–provider and provider–physician–provider.

### 5.3 Protocol

All providers' claims are observed and define the graph; labels of validation and test providers are hidden. This matches an audit in which claims are available for every provider but only some providers have been investigated. Providers are assigned to five stratified folds; in rotation $k$, fold $k$ is the test set, fold $k+1$ the validation set and the remaining three folds the training set, so every provider is tested exactly once. Hyperparameters are selected once, with the same budget of six configurations per method, on a separate stratified 60/20/20 split (seed 1000), and then fixed for the five rotations. Graph models are trained full-batch for at most 100 epochs with early stopping (patience 10) on validation AP. Metrics are AP, ROC-AUC, F1 at the validation-selected threshold, and recall among the top 10% of providers by score (the label prevalence is 9.4%). Methods are compared with paired $t$-tests over the five rotations (Holm-adjusted); with five rotations these tests have limited power and are interpreted together with effect sizes.

## 6 Results on the Controlled Benchmark

Each test period contains on average 4,924 claims, of which 384 belong to four unseen rings and 133 are isolated fraud, so the chance levels of ring AP and isolated AP are 0.080 and 0.029. Table 1 reports ring AP for every method and regime, Fig. 3 shows the main methods with 95% intervals, and Fig. 4 summarises paired differences. Table 2 pools the paired differences over all 54 worlds.

### 6.1 Claim features alone do not reveal rings

When ring claims are drawn like legitimate claims ($\delta = 0$), models that see only the claim's own attributes perform at chance (GB-C 0.084–0.087, MLP-C 0.087–0.095). As $\delta$ grows their ring AP rises to about 0.33 ($\delta = 1$) and 0.81 ($\delta = 2$). This confirms that the benchmark separates the two sources of evidence as intended.

**Table 1** Ring-fraud detection on unseen test rings: average precision of ring versus legitimate claims (chance level 0.08), mean over six worlds per regime. Bold: best per regime. Standard deviations are in the supplementary results files.

| | $\delta = 0$ | | | $\delta = 1$ | | | $\delta = 2$ | | |
|---|---|---|---|---|---|---|---|---|---|
| Method / $s$ | 0.1 | 0.5 | 0.9 | 0.1 | 0.5 | 0.9 | 0.1 | 0.5 | 0.9 |
| GB-C | 0.084 | 0.086 | 0.087 | 0.329 | 0.326 | 0.334 | 0.805 | 0.818 | 0.809 |
| MLP-C | 0.088 | 0.087 | 0.095 | 0.353 | 0.358 | 0.360 | 0.803 | 0.817 | 0.808 |
| LR-CA | 0.923 | 0.888 | 0.868 | 0.968 | 0.958 | 0.952 | 0.988 | 0.986 | 0.987 |
| RF-CA | 0.875 | 0.838 | 0.844 | 0.957 | 0.924 | 0.941 | 0.985 | 0.983 | 0.986 |
| GB-CA | 0.925 | 0.905 | 0.889 | 0.966 | 0.938 | 0.944 | 0.985 | 0.986 | 0.989 |
| GB-CAG | 0.930 | 0.983 | **0.990** | 0.960 | 0.976 | 0.992 | 0.990 | 0.995 | 0.994 |
| SAGE | 0.938 | 0.963 | 0.976 | 0.973 | 0.986 | 0.991 | 0.994 | 0.993 | 0.994 |
| RGCN | 0.933 | 0.953 | 0.954 | 0.970 | 0.982 | 0.986 | **0.995** | 0.994 | 0.996 |
| HAN | 0.908 | 0.906 | 0.921 | 0.957 | 0.964 | 0.969 | 0.988 | 0.989 | 0.989 |
| Simple-HGN | 0.915 | 0.960 | 0.957 | 0.966 | 0.980 | 0.982 | 0.992 | 0.992 | 0.993 |
| HGT | 0.910 | 0.948 | 0.949 | 0.959 | 0.976 | 0.984 | 0.994 | 0.995 | 0.996 |
| SAGE+AG | **0.973** | **0.985** | 0.983 | 0.972 | 0.984 | **0.995** | 0.994 | **0.996** | **0.997** |
| HGT+AG | 0.964 | 0.984 | 0.990 | **0.974** | **0.986** | 0.995 | 0.994 | 0.994 | 0.996 |
| Overlap score | 0.254 | 0.937 | 0.985 | 0.296 | 0.922 | 0.989 | 0.341 | 0.932 | 0.981 |

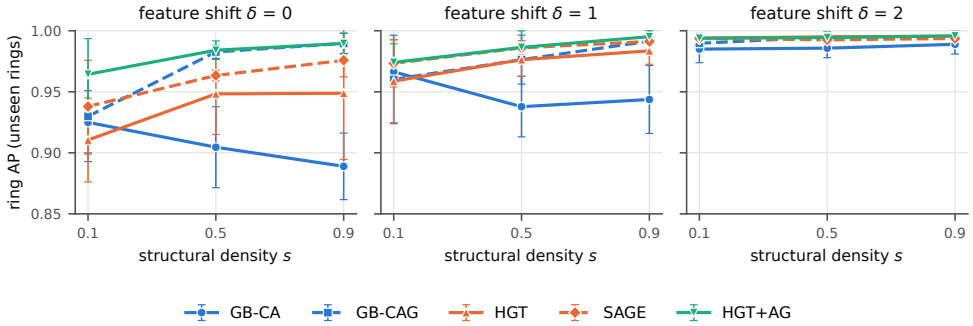



**Fig. 3** Ring AP on unseen test rings by feature shift $\delta$ and structural density $s$ (mean and 95% $t$-interval over six worlds). GB-CA: gradient boosting on claim and aggregate features; GB-CAG: plus engineered graph features; HGT+AG: HGT whose claim nodes also receive the A and G features. Feature-only models (GB-C, MLP-C) lie far below the plotted range (Table 1).

## 6.2 Most ring evidence is captured by entity aggregates

Adding per-period aggregates of the claim's patient, doctor and hospital (GB-CA) raises ring AP at $\delta = 0$ to 0.889–0.925. Ring patients file more claims and see more doctors than typical patients, and ring doctors file bursts of claims; these degree and volume signals are present in every regime by design and are captured by simple counts. The remaining gap is where coordination structure matters.

## 6.3 Coordination structure helps only when the ring block is dense

Engineered graph features add little at low density but substantially at moderate and high density: at $\delta = 0$, GB-CAG improves on GB-CA by $+0.005$ ($s = 0.1$, n.s.), $+0.078$ ($s = 0.5$) and $+0.101$ ($s = 0.9$; Holm $p = 0.003$). The unsupervised overlap score tells the same story: it is weak at $s = 0.1$ (0.254) but reaches 0.937 and 0.985 at $s = 0.5$ and 0.9 without any training. GNNs show the same pattern: SAGE improves on GB-CA by $+0.087$ at $\delta = 0$, $s = 0.9$ (Holm $p = 0.003$). Pooled over all worlds, every graph-aware method except HAN outperforms GB-CA (GB-CAG $+0.031$, SAGE $+0.031$, RGCN $+0.026$, Simple-HGN $+0.024$, HGT $+0.021$, hybrids $+0.039$; all Holm $p < 0.001$; HAN $+0.007$, n.s.). When ring claims are strongly anomalous ($\delta = 2$), all methods that see claim and aggregate features exceed 0.98 and the differences become negligible.

## 6.4 GNNs do not outperform gradient boosting with graph features

The central comparison is between message passing and the same structural information supplied as engineered features. Pooled over the 54 worlds, heterogeneous GraphSAGE matches GB-CAG ($\Delta = -0.0001$, 95% CI $[-0.005, 0.005]$), RGCN is not significantly different ($-0.005$), and Simple-HGN ($-0.008$, Holm $p = 0.024$), HGT ($-0.011$, $[-0.020, -0.002]$, Holm $p = 0.060$) and HAN ($-0.024$, Holm $p < 0.001$) are lower. Within single regimes, where each test has only six worlds, no GNN is significantly better than GB-CAG in any regime; SAGE, HAN and Simple-HGN are significantly worse at $\delta = 0$, $s = 0.5$, and SAGE and Simple-HGN at $\delta = 0$, $s = 0.9$ (Holm-adjusted within each nine-regime family).

The hybrids, which add message passing on top of the engineered features, gain a small but consistent amount over GB-CAG (SAGE+AG $+0.008$ $[0.003, 0.013]$, Holm $p = 0.013$; HGT+AG $+0.008$ $[0.003, 0.013]$, Holm $p = 0.019$), largest in the sparse-structure, no-feature-signal regime ($\delta = 0$, $s = 0.1$: SAGE+AG $+0.043$, unadjusted $p = 0.011$, Holm $p = 0.61$). Message passing therefore carries a little information beyond our engineered features, but in this benchmark it does not replace them.

## 6.5 Message passing dilutes claim-level anomalies

Isolated fraud shows the opposite pattern (Table 3). Every GNN is significantly worse than GB-CAG, by 0.085 (SAGE) to 0.147 (HAN) AP pooled (all Holm $p < 0.001$), with HAN and HGT the weakest. The MLP control, which receives exactly the claim inputs of the GNNs but no messages, comes within 0.019 of GB-CAG. Aggregating a claim's representation with those of its neighbours, which are overwhelmingly legitimate, blurs the claim's own anomaly; the hybrids, whose claim nodes also see the engineered ratios, recover little of the gap. On all fraud combined (overall AP), GB-CAG is better than every pure GNN (by 0.007–0.030, all Holm $p < 0.01$) and indistinguishable from the hybrids.

**Table 2** Paired difference in average precision versus GB-CAG, pooled over all 54 worlds: mean Δ ± half-width of the 95% $t$-interval, and Holm-adjusted paired $t$-test $p$ (within each metric).

| | Ring AP | | Isolated AP | | Overall AP | |
|---|---|---|---|---|---|---|
| Method | Δ± CI | $p$ | Δ± CI | $p$ | Δ± CI | $p$ |
| SAGE | 0.000 ± 0.005 | 0.957 | −0.085 ± 0.019 | <0.001 | −0.007 ± 0.004 | 0.009 |
| RGCN | −0.005 ± 0.007 | 0.277 | −0.106 ± 0.021 | <0.001 | −0.013 ± 0.005 | <0.001 |
| HAN | −0.024 ± 0.010 | <0.001 | −0.147 ± 0.024 | <0.001 | −0.030 ± 0.008 | <0.001 |
| Simple-HGN | −0.008 ± 0.006 | 0.024 | −0.094 ± 0.023 | <0.001 | −0.014 ± 0.005 | <0.001 |
| HGT | −0.011 ± 0.009 | 0.060 | −0.118 ± 0.023 | <0.001 | −0.018 ± 0.007 | <0.001 |
| SAGE+AG | +0.008 ± 0.005 | 0.013 | −0.078 ± 0.015 | <0.001 | 0.000 ± 0.003 | 0.930 |
| HGT+AG | +0.008 ± 0.005 | 0.019 | −0.112 ± 0.018 | <0.001 | −0.002 ± 0.003 | 0.388 |
| MLP-C | −0.560 ± 0.079 | <0.001 | −0.019 ± 0.015 | 0.015 | −0.402 ± 0.062 | <0.001 |

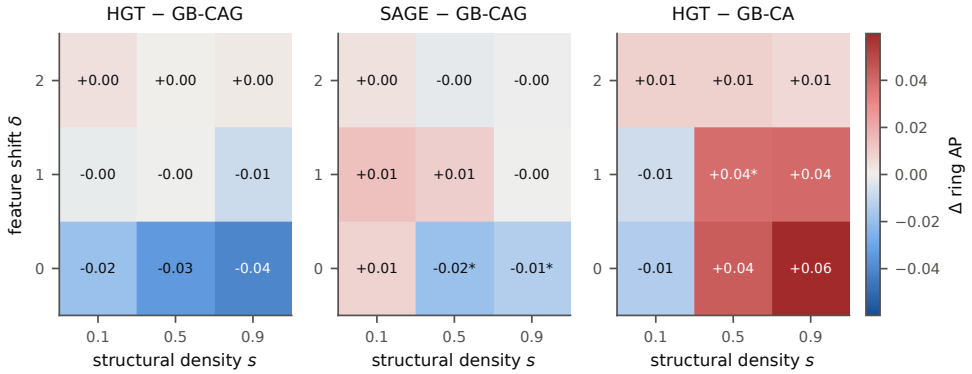



**Fig. 4** Mean paired difference in ring AP (six worlds per cell). Left and middle: HGT and SAGE versus gradient boosting with engineered graph features. Right: HGT versus gradient boosting without graph features. *Holm-adjusted $p < 0.05$ within the nine-regime family.

## 6.6 Routing claims to a GNN adds little beyond combining scores

Table 4 and Fig. 5 evaluate two-stage routing (Section 4.4) on the same test claims as the single models. In the design usually proposed, GB-CAG filters and the GNN decides (re-rank). Validation mostly selected small escalation fractions (5% in 30 of 54 worlds for HGT), and the routed system matched GB-CAG (GB-CAG→HGT $\Delta\text{AP} = -0.001$, 95% CI $[-0.004, 0.002]$; GB-CAG→SAGE +0.001). Escalating more claims to HGT lowered both ring AP (from 0.980 at 5% to 0.968 when every claim is escalated) and, more steeply, isolated AP (from 0.626 to 0.508; Fig. 5b–d), because HGT ranks both less well than GB-CAG. Ordering escalated claims by the mean of the two models' percentiles did better: GB-CAG→HGT gained +0.003 (Holm $p = 0.18$) and GB-CAG→SAGE +0.006 $[0.004, 0.009]$ (Holm $p < 0.001$), on ring (+0.006) and isolated fraud (+0.023) alike. Validation, however, chose to fuse *every* claim in 22

**Table 3** Isolated-fraud detection: average precision of isolated-fraud versus legitimate claims (chance level 0.03), mean over six worlds per regime. Bold: best per regime.

| | $\delta = 0$ | | | $\delta = 1$ | | | $\delta = 2$ | | |
|---|---|---|---|---|---|---|---|---|---|
| Method / $s$ | 0.1 | 0.5 | 0.9 | 0.1 | 0.5 | 0.9 | 0.1 | 0.5 | 0.9 |
| GB-C | 0.471 | 0.473 | 0.477 | 0.560 | 0.556 | 0.557 | 0.644 | 0.649 | 0.633 |
| MLP-C | 0.542 | 0.539 | 0.545 | **0.624** | **0.614** | 0.623 | 0.651 | **0.667** | 0.656 |
| LR-CA | 0.375 | 0.313 | 0.292 | 0.556 | 0.549 | 0.526 | **0.672** | 0.665 | **0.662** |
| RF-CA | 0.551 | 0.506 | 0.469 | 0.586 | 0.560 | 0.547 | 0.626 | 0.608 | 0.604 |
| GB-CA | 0.502 | 0.496 | 0.477 | 0.586 | 0.525 | 0.540 | 0.643 | 0.611 | 0.627 |
| GB-CAG | **0.575** | **0.613** | **0.628** | 0.604 | 0.605 | **0.629** | 0.660 | 0.662 | 0.655 |
| SAGE | 0.447 | 0.448 | 0.509 | 0.536 | 0.548 | 0.555 | 0.615 | 0.605 | 0.602 |
| RGCN | 0.428 | 0.425 | 0.474 | 0.499 | 0.516 | 0.512 | 0.609 | 0.602 | 0.607 |
| HAN | 0.375 | 0.371 | 0.408 | 0.472 | 0.467 | 0.466 | 0.590 | 0.576 | 0.581 |
| Simple-HGN | 0.408 | 0.401 | 0.484 | 0.537 | 0.518 | 0.534 | 0.638 | 0.639 | 0.625 |
| HGT | 0.344 | 0.379 | 0.455 | 0.500 | 0.498 | 0.537 | 0.622 | 0.626 | 0.605 |
| SAGE+AG | 0.457 | 0.503 | 0.586 | 0.493 | 0.548 | 0.582 | 0.582 | 0.584 | 0.598 |
| HGT+AG | 0.378 | 0.454 | 0.553 | 0.444 | 0.498 | 0.566 | 0.549 | 0.576 | 0.604 |
| Overlap score | 0.027 | 0.027 | 0.027 | 0.027 | 0.027 | 0.027 | 0.027 | 0.027 | 0.027 |

(HGT) and 30 (SAGE) of the 54 worlds, and on average 55% and 70% of test claims were escalated: the gain comes from combining two different rankings, not from routing a subset of claims. The feature-level hybrids, which give the GNN the engineered features, improved ring AP (+0.008, Holm $p \leq 0.024$) but lost 0.08–0.11 isolated AP and so matched GB-CAG overall. At the ring level every method was equivalent: in all 54 worlds each test ring had at least one claim among the top 5% of claims, and precision in that budget was 0.989–0.998 for every method.

**Table 4** Two-stage routing versus single models and feature-level hybrids, pooled over the 54 worlds (same test claims for every row). Routing escalates claims whose GB-CAG score exceeds a threshold set on validation data; the escalation fraction is chosen by validation AP. P@5%: precision among the top 5% of test claims. Esc.: mean share of test claims escalated. ΔAP: paired difference in AP from GB-CAG (± half-width of the 95% $t$-interval), Holm-adjusted $p$ over the rows.

| Method | AP | Ring AP | Isol. AP | P@5% | Esc. | ΔAP ($p$) |
|---|---|---|---|---|---|---|
| GB-CAG | 0.947 | 0.979 | 0.626 | 0.996 | – | – |
| HGT | 0.929 | 0.968 | 0.507 | 0.989 | – | −0.018 ± 0.007 (<0.001) |
| SAGE | 0.940 | 0.979 | 0.541 | 0.995 | – | −0.007 ± 0.004 (0.018) |
| HGT+AG | 0.945 | 0.986 | 0.514 | 0.996 | – | −0.002 ± 0.003 (0.776) |
| SAGE+AG | 0.947 | 0.987 | 0.548 | 0.996 | – | 0.000 ± 0.003 (0.970) |
| GB-CAG→HGT, re-rank | 0.945 | 0.979 | 0.606 | 0.995 | 0.16 | −0.001 ± 0.003 (0.970) |
| GB-CAG→HGT, fused | 0.950 | 0.981 | 0.640 | 0.997 | 0.55 | +0.003 ± 0.003 (0.181) |
| GB-CAG→SAGE, re-rank | 0.948 | 0.983 | 0.595 | 0.997 | 0.24 | +0.001 ± 0.003 (0.970) |
| GB-CAG→SAGE, fused | 0.953 | 0.985 | 0.648 | 0.998 | 0.70 | +0.006 ± 0.002 (<0.001) |

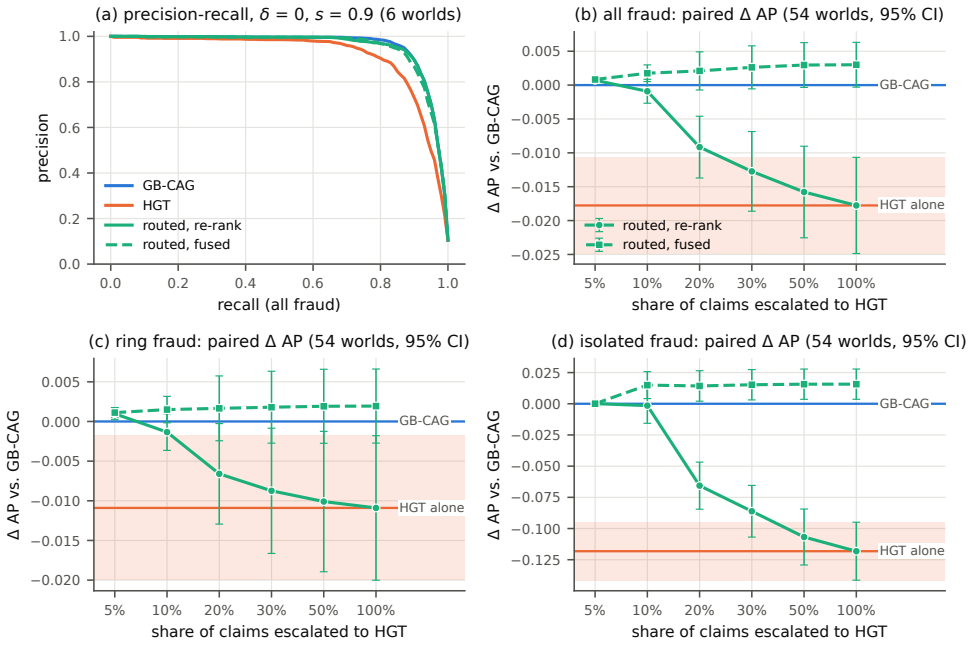



**Fig. 5** Two-stage routing from GB-CAG to HGT. (a) Precision–recall curves for all fraud at $\delta = 0$, $s = 0.9$ (precision interpolated at fixed recall and averaged over six worlds; routed systems at the escalation fraction chosen on validation). (b)–(d) Paired difference in test AP from GB-CAG (zero line) against the share of claims escalated, with 95% $t$-intervals over the 54 worlds; orange line and band: HGT alone. The escalation fraction actually used is chosen on validation; the curves over all fractions are shown only to describe the trade-off.

## 6.7 Cost

Median training time per world (Intel Core i7-1068NG7 laptop CPU, two threads per job with several jobs in parallel, so times are indicative) was 3 s for GB-CAG, 39 s for SAGE, 56 s for RGCN, 86 s for HAN, 126 s for HGT and 144 s for Simple-HGN; scoring a test period took at most 0.25 s for any method. A separate single-thread benchmark on an otherwise idle machine (four worlds at $\delta = 0$) measured scoring directly. For one audit period (about 4,900 claims and 11,200 nodes), building the aggregate and graph features took 0.05 s and the heterogeneous graph 0.01 s, and scoring every claim took 0.05 s with GB-CAG and 0.12 s with HGT. Scoring a single claim, as an online system would, took a median of 2.1 ms (99th percentile 3.1 ms, mostly library-call overhead) with GB-CAG on a precomputed feature row, and 10.7 ms (14.7 ms) with HGT on the claim's exact three-hop computation subgraph (about 126 nodes, 1% of the period graph; extraction 2.8 ms, forward pass 7.7 ms), which reproduces the full-graph score to within $1.2 \times 10^{-7}$. Routing would therefore cut the online GNN cost in proportion to the share of claims escalated. In batch scoring the escalated claims' subgraphs overlap: escalating 10% of the claims already touched 55% of the period graph, and scoring that union took 0.08 s against 0.12 s for the whole period. None of

these figures includes keeping aggregates, graph features and degree counts up to date as claims arrive, which an online system would need and our code does not implement.

# 7 Mechanism and Memorisation

## 7.1 What isolated and ring fraud look like

Fig. 6 shows the test period of one world (seed 1, $\delta = 0$) with the outcomes of GB-CAG and HGT at their validation-selected thresholds. An isolated fraud claim sits on an ordinary patient–doctor edge, so nothing in the graph distinguishes it and only its own attributes can. A ring at $s = 0.1$ links each recruited patient to one or two of the four ring doctors, a pattern that resembles ordinary shared care, whereas at $s = 0.9$ the same members form an almost complete block. The two worlds share the recruited ring members and every legitimate and isolated-fraud claim, and differ only in which ring links exist. In this world GB-CAG flagged 81 and 91 of the ring's 96 claims at $s = 0.1$ and $s = 0.9$, and HGT 62 and 94: the graph model gains most where the block is dense, as in the averages of Table 1. The figure illustrates one world only; the quantitative statements rest on all 54.

## 7.2 What the models need

Tables 5 and 6 and Fig. 7 report ablations in the regime where structure matters most ($\delta = 0$, $s = 0.9$). Degree-preserving rewiring—permuting which doctor bills each claim within a period, which keeps every patient's and doctor's claim volume but destroys the ring's co-occurrence block—reduces every graph-aware method to the same level (0.757–0.770), close to gradient boosting without graph features after rewiring (0.748). The GNNs therefore do exploit coordination structure, and once it is removed nothing distinguishes them from a model that sees volume alone.

Depth matters: with a single message-passing layer, ring AP falls by 0.135 (SAGE) and 0.161 (HGT); two and three layers perform alike. Removing the direct patient–doctor relation costs little ($-0.016$ and $-0.047$, not significant after correction), because two-hop paths through claim nodes carry the same information. The largest effect is specific to HGT: without degree encodings its ring AP drops from 0.949 to 0.569 ($-0.379$, Holm $p = 0.003$), whereas sum-aggregating SAGE changes little ($-0.029$, n.s.). Softmax attention normalises over neighbours and so cannot count them; supplying counts as input features, or aggregating by summation, restores the signal [28].

## 7.3 Which parts of HGT matter

Table 6 changes one factor of the tuned HGT at a time in the same regime. Depth beyond two layers and the number of attention heads made no significant difference (four layers $+0.002$; one, two and eight heads $-0.028$ to $+0.013$; all Holm $p > 0.5$), so there is no sign of over-smoothing up to four layers on these graphs. Among the relations, removing the patient–claim edges cost most ($-0.219$, Holm $p = 0.005$); removing the claim–doctor edges cost 0.113 on rings (Holm $p = 0.085$) and 0.147 on isolated fraud ($p < 0.001$), while hospital nodes and the direct patient–doctor relation

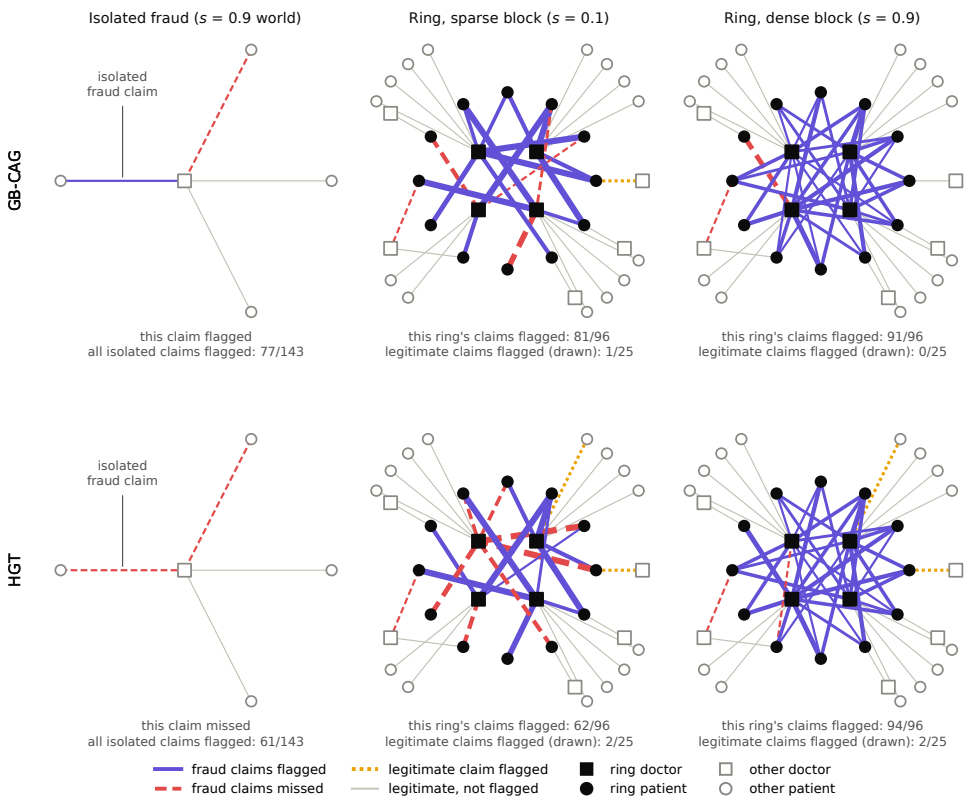



**Fig. 6** Isolated and ring fraud in one benchmark world (seed 1, $\delta = 0$; test period), with model outcomes at each model's validation-selected threshold (top: GB-CAG; bottom: HGT). Nodes carry the ground truth (filled: ring member; squares: doctors; circles: patients); edges are patient–doctor pairs with claims in the period, coloured by the outcome for the majority of their claims (line width: number of claims). Left: an isolated fraud claim drawn at random, its patient's doctors and up to ten other patients of its doctor. Middle and right: one test ring, every doctor its patients visit and up to four other patients per ring doctor, at low and high structural density; the two worlds share all recruited members and all legitimate claims. The figure illustrates the generator's typologies; it is not evidence that real rings look alike.

contributed nothing measurable. The node inputs separate the two tasks cleanly. With claim attributes set to zero, ring AP was unchanged ($-0.012$) and isolated AP fell to chance (0.036); an HGT that saw only the degree encodings and the graph reached a ring AP of $0.982 \pm 0.018$, as high as the full model, and detected no isolated fraud. HGT's ring detection is therefore structural—counts of who bills whom, propagated along patient–claim–doctor paths—and its claim attributes serve isolated fraud only. This is the information that the engineered shared-patient features summarise directly. HGT also varied between worlds (standard deviations up to 0.11), so with six worlds differences smaller than about 0.05 cannot be resolved.

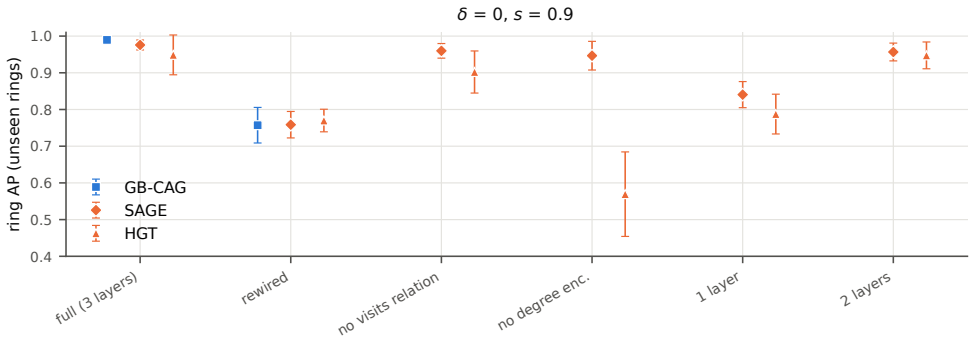



**Fig. 7** Mechanism ablations at $\delta = 0$, $s = 0.9$ (mean and 95% $t$-interval over six worlds). Rewiring preserves every node's degree but destroys the ring block.

**Table 5** Mechanism ablations at $\delta = 0$, $s = 0.9$ (six worlds): ring AP (mean ± s.d.) and paired change from the full model, with Holm-adjusted $p$ over every ablation comparison in this table and Table 6. The tuned GNNs use three layers; HGT's single-factor variants are in Table 6.

| Method | Variant | Ring AP | Δ vs. full |
|---|---|---|---|
| GB-CA | Full model | 0.889 ± 0.026 | – |
| | Degree-preserving rewiring | 0.748 ± 0.027 | −0.141 ($p = 0.018$) |
| | + ring identifier (legacy leak) | 1.000 ± 0.000 | +0.111 ($p = 0.003$) |
| GB-CAG | Full model | 0.990 ± 0.008 | – |
| | Degree-preserving rewiring | 0.757 ± 0.046 | −0.232 ($p = 0.002$) |
| SAGE | Full model | 0.976 ± 0.013 | – |
| | Degree-preserving rewiring | 0.759 ± 0.034 | −0.217 ($p = 0.002$) |
| | No patient–doctor relation | 0.960 ± 0.019 | −0.016 ($p = 0.360$) |
| | No degree encodings | 0.947 ± 0.037 | −0.029 ($p = 1.000$) |
| | 1 message-passing layer | 0.841 ± 0.034 | −0.135 ($p = 0.002$) |
| | 2 layers | 0.957 ± 0.023 | −0.019 ($p = 0.413$) |
| HGT | Full model | 0.949 ± 0.052 | – |
| | Degree-preserving rewiring | 0.770 ± 0.029 | −0.179 ($p = 0.036$) |
| | No reverse relations (legacy graph) | 0.772 ± 0.024 | −0.177 ($p = 0.022$) |

## 7.4 Which inputs the trained models use

To see which inputs the trained models use, without retraining, we permuted one input group at a time across the test period, five times each, and recorded the drop in ring AP: for GB-CAG the C, A and G features and each G feature separately, and for the GNNs the claim attributes, static entity attributes and degree encodings across nodes of each type. At $\delta = 0$, GB-CAG relied on aggregates when rings were sparse (A: −0.80 at $s = 0.1$; G: −0.13) and switched to the graph features as the block became dense (G: −0.81 and −0.88 at $s = 0.5$ and 0.9; A: below 0.06), above all the normalised triadic closure of the claim (−0.47 on its own at $s = 0.9$). SAGE and HGT relied almost entirely on degree encodings (−0.76 to −0.89 at every density) and hardly at all on

**Table 6** HGT single-factor ablations at $\delta = 0$, $s = 0.9$ (six worlds; one factor changed at a time from the tuned configuration: 3 layers, 4 heads, hidden size 32). Ring AP (mean ± s.d.), isolated AP (mean) and paired change from the full model; Holm-adjusted $p$ over every ablation comparison in this table and Table 5.

| Variant | Ring AP | Δ ring AP | Isol. AP | Δ isol. AP |
|---|---|---|---|---|
| Full model | 0.949 ± 0.052 | – | 0.455 | – |
| *Depth (full: 3)* | | | | |
| 1 layer | 0.788 ± 0.052 | −0.161 ($p = 0.020$) | 0.239 | −0.216 ($p = 0.007$) |
| 2 layers | 0.947 ± 0.035 | −0.001 ($p = 1.000$) | 0.423 | −0.032 ($p = 1.000$) |
| 4 layers | 0.951 ± 0.043 | +0.002 ($p = 1.000$) | 0.461 | +0.005 ($p = 1.000$) |
| *Heads (full: 4)* | | | | |
| 1 attention head | 0.920 ± 0.049 | −0.028 ($p = 1.000$) | 0.384 | −0.071 ($p = 0.228$) |
| 2 heads | 0.911 ± 0.081 | −0.037 ($p = 0.506$) | 0.473 | +0.018 ($p = 0.813$) |
| 8 heads | 0.961 ± 0.023 | +0.013 ($p = 1.000$) | 0.524 | +0.069 ($p = 0.282$) |
| *Relations* | | | | |
| No patient–claim relation | 0.730 ± 0.055 | −0.219 ($p = 0.005$) | 0.311 | −0.144 ($p = 0.333$) |
| No claim–doctor relation | 0.836 ± 0.037 | −0.113 ($p = 0.085$) | 0.308 | −0.147 ($p < 0.001$) |
| No patient–doctor relation | 0.902 ± 0.055 | −0.047 ($p = 0.413$) | 0.423 | −0.032 ($p = 1.000$) |
| No hospital nodes | 0.946 ± 0.032 | −0.002 ($p = 1.000$) | 0.427 | −0.028 ($p = 1.000$) |
| *Node inputs* | | | | |
| No degree encodings | 0.569 ± 0.110 | −0.379 ($p = 0.003$) | 0.375 | −0.080 ($p = 0.813$) |
| Claim attributes zeroed | 0.937 ± 0.064 | −0.012 ($p = 1.000$) | 0.036 | −0.419 ($p < 0.001$) |
| Entity attributes zeroed | 0.969 ± 0.027 | +0.020 ($p = 1.000$) | 0.444 | −0.011 ($p = 1.000$) |
| Degree encodings only | 0.982 ± 0.018 | +0.033 ($p = 1.000$) | 0.040 | −0.415 ($p < 0.001$) |

claim or entity attributes (below 0.05). The two kinds of score are not on a common scale: GB-CAG's measure reliance on engineered summaries of the graph, whereas a GNN's degree encodings are raw counts whose relational meaning is extracted by message passing, and permutation understates groups whose information is duplicated elsewhere. Read together with the ablations, they are consistent with both model families detecting dense rings from the same evidence: how many counterparties a claim's patient and doctor have, and how many of them they share. (Full results, including isolated fraud and test-time rewiring, are in the repository.)

## 7.5 Two evaluation pitfalls, quantified

The mechanism ablations (Table 5) also quantify two defects that are easy to introduce. Adding the generator's ring identifier to the feature set—a column that is non-zero only for ring claims—raises gradient boosting from 0.889 to a perfect 1.000. Building the heterogeneous graph with one-directional relations only (patient→claim→doctor→hospital) leaves claim nodes reachable only from patients, so doctor context never reaches them; HGT then loses 0.177 AP. Both defects produce plausible-looking numbers and neither is caught by aggregate metrics alone; the tests listed in Section 10 detect both.

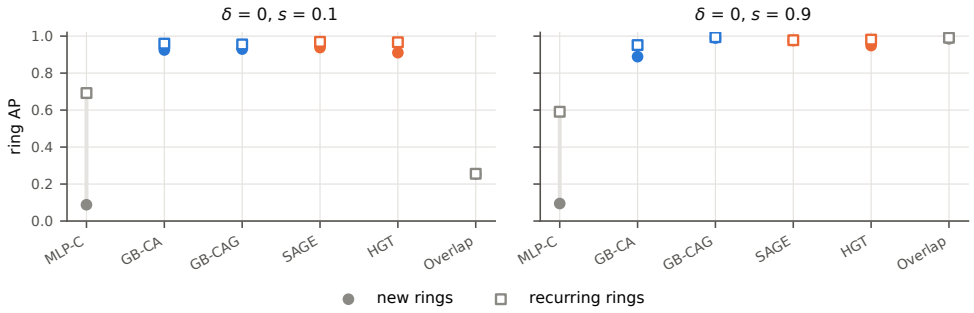



**Fig. 8** Ring AP on new versus recurring rings ($\delta = 0$, six worlds).

## 7.6 Recognising known members is not detecting rings

Table 7 and Fig. 8 compare new rings (emergent protocol) with rings whose members recur from training (persistent protocol). The MLP control, which sees only the claim and the static attributes of its patient, doctor and hospital, rises from chance to 0.692 ($s = 0.1$) and 0.591 ($s = 0.9$) when rings recur: it identifies the members from their attributes. HGT gains 0.056 at $s = 0.1$ (95% CI $[0.029, 0.084]$), and gradient boosting with aggregates gains 0.035–0.063, while the unsupervised overlap score, which cannot memorise, is unchanged. Evaluations in which test rings share members with training rings therefore overstate how well a model detects coordination, by an amount that differs between methods and can reorder them.

**Table 7** Ring AP when test rings are new (emergent) versus recurring members of training rings (persistent), $\delta = 0$, six worlds each; gain = persistent − emergent (paired over worlds, 95% $t$-interval). The overlap score is unsupervised and cannot memorise.

| | $s = 0.1$ | | | $s = 0.9$ | | |
|---|---|---|---|---|---|---|
| Method | new | recurring | gain | new | recurring | gain |
| MLP-C | 0.088 | 0.692 | +0.604 [+0.523, +0.685] | 0.095 | 0.591 | +0.496 [+0.416, +0.577] |
| GB-CA | 0.925 | 0.959 | +0.035 [+0.014, +0.055] | 0.889 | 0.952 | +0.063 [+0.034, +0.092] |
| GB-CAG | 0.930 | 0.955 | +0.025 [−0.012, +0.062] | 0.990 | 0.994 | +0.005 [−0.002, +0.012] |
| SAGE | 0.938 | 0.969 | +0.031 [−0.019, +0.081] | 0.976 | 0.978 | +0.002 [−0.015, +0.020] |
| HGT | 0.910 | 0.967 | +0.056 [+0.029, +0.084] | 0.949 | 0.981 | +0.032 [−0.017, +0.082] |
| Overlap score | 0.254 | 0.256 | +0.002 [−0.137, +0.142] | 0.985 | 0.991 | +0.006 [−0.009, +0.020] |

# 8 Results on Real Data

Table 8 and Fig. 9 report provider-level results over the five rotations (1,082 test providers each, about 101 fraudulent). Gradient boosting on provider aggregates (GB-P) has the highest AP ($0.704 \pm 0.045$). Adding label-free graph features (GB-PG, $-0.009$, 95% CI $[-0.029, 0.011]$) or graph and label-neighbourhood features (GB-PGL, $-0.007$ $[-0.021, 0.007]$) does not improve it. Every GNN has lower AP than GB-P, by $-0.029$ (HGT) to $-0.083$ (RGCN); with five rotations most of these differences are not significant after Holm adjustment over all 24 comparisons, the exception being SAGE+G ($-0.051$, Holm $p = 0.021$). The hybrids, which give the GNN the provider and graph features, are no better than the plain GNNs. ROC-AUC lies between 0.927 and 0.942 for all methods and recall among the top 10% of providers between 0.59 and 0.64, so the methods rank providers similarly; the AP differences come from the top of the ranking. Under the same parallel load as before, median training time per rotation was 2–15 minutes for the GNNs (range 1.4–24 minutes) against 7 s for GB-P. Because the tuning split shares providers with the test folds, we re-selected the configuration of every tabular and MLP model inside each rotation, using only that rotation's validation fold. AP changed by at most 0.010 (GB-P 0.706 versus 0.704; no paired difference significant), and GB-P remained the best method, so the overlap does not explain the ranking.

**Table 8** Provider-level fraud detection on the public Medicare-style provider dataset (5-fold rotation; mean ± s.d. over folds). F1 at the validation-selected threshold; Recall@10% = share of fraudulent providers among the top 10% scored.

| Method | AP | ROC-AUC | F1 | Recall@10% |
|---|---|---|---|---|
| LR-P | 0.678 ± 0.041 | 0.940 ± 0.017 | 0.617 ± 0.030 | 0.644 ± 0.023 |
| RF-P | 0.681 ± 0.049 | 0.942 ± 0.017 | 0.624 ± 0.031 | 0.644 ± 0.025 |
| **GB-P** | 0.704 ± 0.045 | 0.939 ± 0.020 | 0.624 ± 0.028 | 0.636 ± 0.027 |
| GB-PG | 0.695 ± 0.040 | 0.937 ± 0.019 | 0.600 ± 0.043 | 0.642 ± 0.024 |
| GB-PGL | 0.697 ± 0.042 | 0.937 ± 0.019 | 0.595 ± 0.045 | 0.634 ± 0.019 |
| MLP-P | 0.685 ± 0.052 | 0.940 ± 0.019 | 0.613 ± 0.028 | 0.642 ± 0.032 |
| SAGE | 0.664 ± 0.047 | 0.938 ± 0.018 | 0.605 ± 0.036 | 0.624 ± 0.046 |
| RGCN | 0.621 ± 0.054 | 0.927 ± 0.019 | 0.578 ± 0.029 | 0.593 ± 0.029 |
| HAN | 0.666 ± 0.046 | 0.938 ± 0.018 | 0.583 ± 0.028 | 0.636 ± 0.045 |
| Simple-HGN | 0.670 ± 0.057 | 0.939 ± 0.020 | 0.600 ± 0.037 | 0.630 ± 0.036 |
| HGT | 0.676 ± 0.058 | 0.939 ± 0.019 | 0.613 ± 0.051 | 0.640 ± 0.051 |
| SAGE+G | 0.653 ± 0.048 | 0.937 ± 0.018 | 0.612 ± 0.040 | 0.623 ± 0.031 |
| HGT+G | 0.648 ± 0.032 | 0.933 ± 0.014 | 0.589 ± 0.022 | 0.615 ± 0.035 |

The graph statistics explain why. Almost every provider (99.5%) shares beneficiaries with other providers (median 30 co-providers), but labels are not assortative through this relation: on average 23.9% of a fraudulent provider's co-providers are labelled fraudulent, against 25.3% for a legitimate provider (both well above the 9.4% base rate because shared beneficiaries concentrate on large providers). Shared physicians show mild assortativity (19.1% versus 14.2%) but connect only 73% of providers, with a median of two co-providers. Fraudulent providers are, however, much larger

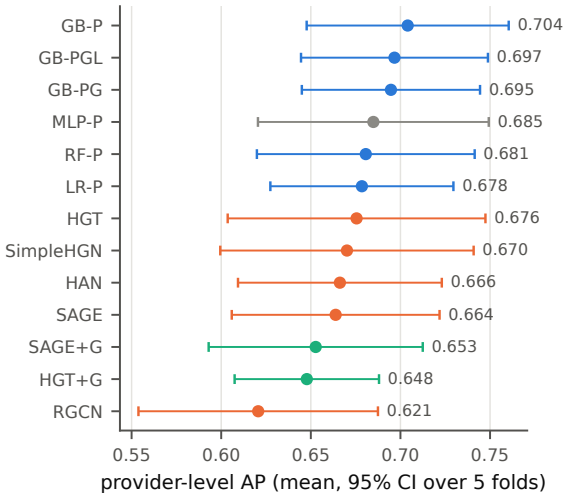



**Fig. 9** Provider-level AP on the real data (mean and 95% $t$-interval over five rotations). Blue: tabular; orange: GNN; aqua: hybrid; gray: MLP.

(median 156 versus 27 claims), a volume signal that provider aggregates capture directly. In the terms of our benchmark, these data resemble the low-density regime, where structure added nothing beyond per-entity aggregates. The labels mark potentially fraudulent providers without identifying rings, so the absence of a graph benefit here does not show that rings are absent from Medicare data; it shows that this public labelling contains no exploitable relational signal for the models tested.

# 9 Discussion

## 9.1 When does relational structure help?

Three conditions emerge from the controlled study. First, coordination must leave a structural trace beyond volume: at low block density, rings were detected mainly through the elevated utilisation of their members, which per-entity counts capture. Second, the trace must be dense enough to distinguish a ring from ordinary shared care: graph information added 0.04–0.10 AP at moderate and high density (most when ring claims looked legitimate) but nothing measurable at low density. Third, the claims must not already be anomalous on their own: when ring claims were strongly upcoded ($\delta = 2$), every model with claim and aggregate features exceeded 0.98 AP and structure became redundant.

## 9.2 Why GNNs did not beat engineered graph features

When structure did help, gradient boosting with a handful of shared-patient statistics captured it as well as or better than the best GNN. The mechanism analyses suggest why. The useful signal here is largely *counting*—how many of a patient's doctors also share other patients—and attention- or mean-based aggregators cannot count: HGT

lost most of its ring detection without degree encodings, whereas sum-aggregating GraphSAGE barely changed, and an HGT given only degree encodings and the graph detected rings as well as the full model (Section 7). This mirrors the expressiveness argument for sum aggregation [28]. GNNs also dilute claim-level anomalies through neighbourhood averaging, which made them markedly worse on isolated fraud, and they readily memorise the attributes of training-ring members, which inflates performance when rings recur but does not transfer to new rings. Hybrids that combine message passing with engineered features recovered a small, consistent gain over gradient boosting on ring fraud, indicating that the GNN extracted some structure our features miss, at 9–40 times the training cost.

These findings are consistent with evidence from other domains that tree ensembles with neighbourhood features are strong graph anomaly detectors [7, 8], and extend it to heterogeneous patient–provider claim graphs with controlled typologies.

### 9.3 Why a small fixed benchmark can point the other way

To show how a small, fixed benchmark can mislead, we re-analysed one: a synthetic dataset of 2,100 claims with five injected rings, produced by a simpler generator that is included in our repository, evaluated with its own date-sorted 60/20/20 split (Table 9). Such a dataset appears to test ring detection, but this one cannot tell whether relational structure helps, for three reasons. First, its ring claims were drawn from the same inflated distribution as its isolated fraud, so they were anomalous on their own: claim amount alone separates ring from legitimate claims with ROC-AUC 0.94, more than in our most anomalous regime ($\delta = 2$, 0.90), and gradient boosting on three raw claim attributes ranks all 21 test ring claims above every legitimate claim (ring AP 1.000). Second, the same five rings were active for all 24 months, so every test ring claim was billed by a doctor with labelled ring claims in training, which is the memorisation setting of Section 7. Third, the test split held 21 ring claims (3–6 per ring): one claim moves ring recall by 4.8 points, even a perfect recall has an exact 95% interval of $[0.84, 1.00]$, so a recall gap of a few claims between two models is within noise, and a single dataset says nothing about variation between datasets. None of these properties is peculiar to that dataset. A small, fixed benchmark whose injected rings differ from legitimate claims in their features, recur across the split and contribute a few dozen test claims cannot separate relational learning from claim-level anomaly detection or from recognising known members. Controlling the feature-level signal, testing on new rings and replicating worlds removes all three confounds. Because the generator, features, evaluation unit and tuning differ, numbers from the two set-ups are not directly comparable.

### 9.4 Real data

On the public provider-labelled data, gradient boosting on provider aggregates was the strongest method, and neither engineered graph features, label-neighbourhood features nor any GNN improved on it. The labels are not assortative through shared beneficiaries, and fraudulent providers differ mainly in volume, so the data sit in the regime where our benchmark also found no benefit from structure. This is consistent

**Table 9** A small fixed benchmark (2,100 claims, five rings; simpler generator in the repository), re-analysed with its own split (date-sorted, 60/20/20), compared with our benchmark. Benchmark ranges span $\delta = 0$ to $\delta = 2$ ($s = 0.5$, six worlds each; ring AP and recall of GB-C pooled over $s$). Shift and ROC-AUC use raw claim amounts, not adjusted for specialty, for both. Interval: exact 95% binomial.

| | Fixed dataset | This benchmark |
|---|---|---|
| Claims in total (in the test split / period) | 2,100 (420) | 20,176 (4,924), mean per world |
| Ring claims in test | 21 (3–6 per ring) | 384 (4 rings × 96) |
| Rings | 5, active all 24 months | 4 per period, each active 6 months |
| Test-ring doctors with labelled ring claims in training | 100% | 0% (by design) |
| Ring vs. legitimate log-amount shift (s.d.) | 1.49 | 0.11 ($\delta = 0$) to 1.87 ($\delta = 2$) |
| ROC-AUC of claim amount alone, ring vs. legitimate | 0.94 | 0.53 to 0.90 |
| Ring AP, GB on claim attributes only | 1.000 | 0.086 to 0.811 |
| Ring recall, GB on claim attributes only | 21/21 [0.84, 1.00] | 0.15 to 0.76 |
| Independent datasets | 1 | 54 |

with the controlled results but is not an independent confirmation of the dense-ring regime: validating that regime requires real data with labelled collusion, which are not publicly available.

## 9.5 Practical recommendations

- Start with gradient boosting on claim features, per-entity aggregates and a small set of shared-patient graph features; on our benchmark this was never significantly worse than any pure GNN, within 0.01 AP of the hybrids on average, and far cheaper.
- Evaluate on rings (or providers) that do not appear in training, and compare against a no-graph control with identical inputs; otherwise memorisation of known members can masquerade as relational learning.
- If a GNN is used, include reverse relations, verify that information reaches the classified nodes, and give attention-based architectures structural (degree) encodings or prefer sum aggregation.
- Keep isolated-fraud detection on a claim-level model; message passing degraded it consistently.
- Do not expect a GNN second stage to pay for itself. Letting a GNN re-rank the claims that gradient boosting flags gained nothing here; averaging the two models' rankings gained at most 0.006 AP, and in an audit setting, where a whole period is scored at once, routing saves little computation (Section 6).

### 9.6 Limitations

The controlled findings hold for our generator. Its distributions are simple, labels are noise-free and immediate, rings are static during their active period and of a single size, and ring prevalence (about 8% of test-period claims) is far above realistic levels so that each world has enough positives for stable estimates. Ring members also carry a volume signal in every regime, so our regimes vary the *additional* structural evidence rather than all structural evidence. We tuned each method once, on development worlds from a single regime ($\delta = 0.5$, $s = 0.5$); per-regime tuning could change small differences. Six worlds per regime give limited power within a regime, which is why we emphasise pooled comparisons. Graph models were trained full-batch on graphs of about 11,000 nodes per period; larger graphs would need sampling-based training [30]. The architecture and reliance analyses were run in one regime ($\delta = 0$, $s = 0.9$ for the ablations) with one model seed per world, and permutation importance can understate a feature group whose information is duplicated in another; these analyses describe what the trained models use, not causal effects of the generator. Routing was evaluated with one first stage, two GNNs and a grid of six escalation fractions; the GNNs' validation scores come from the period used for early stopping, which slightly favours escalation. Cost was measured on a single laptop CPU thread and excludes the upkeep of aggregates, graph features and degree counts as claims arrive, data retrieval and concurrency, none of which exist in our code. On the real data, labels are per provider and of undocumented provenance, and the observation window is a single year. Hyperparameters were selected on a split whose validation providers overlap the evaluation test folds; re-selecting the tabular and MLP configurations inside each rotation (Section 8) removes this overlap for those models, but the GNNs keep the shared selection, which if anything favours them. Finally, the benchmark does not model adaptive fraudsters, temporal ring dynamics, which temporal graph models target [37], or realistic coding semantics such as those of synthetic health records [38].

## 10 Reproducibility

All results are produced by one released package from raw inputs. Every run writes a record containing the git commit, a hash of its configuration, the hashes of the generated claim and ground-truth tables, the library versions and the per-claim (or per-provider) test predictions; the tables and figures of this paper are generated from these records by a single script. Automated tests, run in continuous integration, check that (i) no ground-truth column reaches any feature matrix and features are invariant to permuting the labels; (ii) a period's features and graph are unchanged when claims of other periods are perturbed; (iii) emergent rings in different splits share no doctor or patient; (iv) at $\delta = 0$ ring claims are distributed like legitimate claims; (v) features, graphs and labels stay aligned by claim identifier; and (vi) for every GNN, perturbing doctor features changes claim logits, so relational information actually reaches the classified nodes; (vii) a routed score depends only on the claim's own two scores and validation statistics; (viii) rewiring the test period leaves the features of other periods unchanged; and (ix) scoring a claim on its $L$-hop computation subgraph reproduces its full-graph score. The original runs were made from a working tree that had not yet

been committed, so their records do not pin the code version; retraining GB-CAG, SAGE and HGT with the released code on all 54 worlds reproduced their stored test predictions (Section 4.4), which ties the reported numbers to the released code. Records now also store a hash of the research sources at process start. A checklist mapping this study to the REFORMS recommendations [39] and a datasheet for the generator [40] are included in the repository.

# 11 Conclusion

We asked when relational structure helps healthcare fraud detection and answered it with a benchmark that varies the feature-level and structural signal of provider rings independently, tests on rings never seen in training, and checks the conclusions on public real data. Structure helped only when rings formed dense doctor–patient blocks, and in that regime a gradient-boosting model with a handful of engineered shared-patient features matched or outperformed five heterogeneous GNNs at a fraction of the cost; adding message passing on top of those features, or routing high-risk claims to a GNN, gained little. GNNs generalised to new rings only with reverse relations, sufficient depth and counting ability, degraded isolated-fraud detection, and could memorise known ring members in ways that inflate conventional evaluations. A small fixed benchmark whose rings are anomalous in their features, recur across the split and contribute few test claims can suggest the opposite conclusion. On provider-labelled real data no graph-based method improved on tabular aggregates. We recommend that graph models for claims fraud be justified against a graph-augmented tabular baseline and a no-graph control, on entities unseen in training. The benchmark, code and tests are released to support such comparisons; the most valuable next step is real data with labelled collusive schemes.

**Supplementary information.** The repository contains the per-world results (including standard deviations, F1, ROC-AUC and recall at a fixed investigation budget for every regime and method), the tuning logs and the test predictions underlying every table and figure.

# Statements and Declarations


- **Funding.** The authors did not receive support from any organization for the submitted work.
- **Competing interests.** The authors declare no competing interests.
- **Ethics approval and consent to participate.** Not applicable: the study uses synthetic data and a public, de-identified dataset released under a CC0 licence; no human participants were involved.
- **Consent for publication.** Not applicable.
- **Data availability.** The synthetic benchmark is generated by the released code from fixed seeds. The real-world data are available from Kaggle [36] under CC0; file checksums are listed in the repository.

- **Code availability.** Code, configurations, tests and result records are available at https://github.com/akshay-hudev/Fraud-Guard (directory `research/`) under the MIT licence. An archived release with a DOI will be provided upon acceptance.
- **Author contribution.** H M Akshay: conceptualization, methodology, software, investigation, formal analysis, data curation, visualization and writing—original draft. Gaurav S B, Eshita Narayana and Harshita Srivastava: methodology, investigation, formal analysis and writing—original draft. Malini M Patil: supervision and writing—review and editing. All authors read and approved the final manuscript.

# Appendix A Generator parameters and hyperparameters

**Table A1** Generator parameters (defaults in `research/synth/generator.py`).

| Parameter | Value |
|---|---|
| Patients / doctors / hospitals / regions / specialties | 6,000 / 300 / 40 / 5 / 8 |
| Audit periods | 4 × 6 months (train: 1–2, validation: 3, test: 4) |
| Legitimate claims per patient | Poisson(gamma rate; mean 3, shape 1.5) over 24 months |
| Share of legitimate claims with a regular doctor | 0.85 (1–3 regular doctors per patient, same region) |
| Claim amount | log-normal: specialty mean + 0.15 (procedures − 1) + 0.8 inpatient, s.d. 0.6 |
| Isolated fraud | 3% of legitimate volume; feature shift 2.0 |
| Ring size | 4 doctors, 12 recruited patients (utilisation-weighted) |
| Ring volume | 16 claims per month during activity; monthly burst days |
| Rings per period | 4 (emergent: new members each period; persistent: same 4 rings) |
| Feature shift $\delta$ | 0, 1, 2 (development worlds: 0.5) |
| Structural density $s$ | 0.1, 0.5, 0.9 (development worlds: 0.5) |
| Worlds per regime | 6 (seeds 1–6); development worlds: seeds 901, 902 |

# References


[1] National Health Care Anti-Fraud Association. The Challenge of Health Care Fraud. Accessed 20 October 2026. Available from: https://www.nhcaa.org/tools-insights/about-health-care-fraud/the-challenge-of-health-care-fraud/.

[2] Schlichtkrull M, Kipf TN, Bloem P, van den Berg R, Titov I, Welling M. Modeling relational data with graph convolutional networks. In: The Semantic Web (ESWC 2018). Lecture Notes in Computer Science, vol. 10843. Springer; 2018. p. 593–607.

**Table A2** Selected hyperparameters (equal budget per method; hybrids inherit their base configuration; methods sharing a search space were offered the same candidate configurations). Synthetic: tuned on development worlds; real data: tuned on a separate split.

| Method | Synthetic benchmark | Real data |
|---|---|---|
| LR | C=100 | C=1 |
| RF | max depth=16; max features=sqrt; min samples leaf=3; n estimators=300 | max depth=16; max features=0.3; min samples leaf=10; n estimators=300 |
| GB (C) | l2=0; learning rate=0.03; max iter=200; max leaf nodes=15; min samples leaf=10 | – |
| GB (C+A / P) | l2=1; learning rate=0.1; max iter=200; max leaf nodes=63; min samples leaf=50 | l2=0; learning rate=0.03; max iter=200; max leaf nodes=15; min samples leaf=50 |
| GB (C+A+G / P+G) | l2=1; learning rate=0.1; max iter=400; max leaf nodes=15; min samples leaf=20 | l2=0; learning rate=0.03; max iter=200; max leaf nodes=15; min samples leaf=50 |
| GB (P+G+L) | – | l2=0; learning rate=0.03; max iter=200; max leaf nodes=15; min samples leaf=50 |
| MLP | dropout=0.3; hidden=128; layers=2; lr=0.001; weight decay=1e-05 | dropout=0.3; hidden=128; layers=2; lr=0.001; weight decay=1e-05 |
| SAGE | aggr=sum; dropout=0.1; hidden=32; layers=3; lr=0.003; weight decay=0.001 | aggr=mean; dropout=0.3; hidden=64; layers=3; lr=0.003; weight decay=0.0001 |
| RGCN | aggr=sum; dropout=0.3; hidden=64; layers=2; lr=0.003; weight decay=0.001 | aggr=sum; dropout=0.3; hidden=64; layers=2; lr=0.01; weight decay=0.001 |
| HAN | dropout=0.3; heads=4; hidden=32; layers=3; lr=0.01; weight decay=1e-05 | dropout=0.1; heads=2; hidden=64; layers=2; lr=0.003; weight decay=0.0001 |
| Simple-HGN | dropout=0.3; heads=4; hidden=64; layers=3; lr=0.003; weight decay=0.0001 | dropout=0.1; heads=4; hidden=64; layers=2; lr=0.01; weight decay=0.001 |
| HGT | dropout=0.3; heads=4; hidden=32; layers=3; lr=0.01; weight decay=1e-05 | dropout=0.1; heads=4; hidden=64; layers=2; lr=0.01; weight decay=0.001 |


[3] Wang X, Ji H, Shi C, Wang B, Ye Y, Cui P, et al. Heterogeneous graph attention network. In: Proceedings of the World Wide Web Conference (WWW); 2019. p. 2022–2032.

[4] Hu Z, Dong Y, Wang K, Sun Y. Heterogeneous graph transformer. In: Proceedings of the Web Conference 2020 (WWW); 2020. p. 2704–2710.

[5] Lv Q, Ding M, Liu Q, Chen Y, Feng W, He S, et al. Are we really making much progress? Revisiting, benchmarking and refining heterogeneous graph neural networks. In: Proceedings of the 27th ACM SIGKDD Conference on Knowledge Discovery and Data Mining; 2021. p. 1150–1160.

[6] Grinsztajn L, Oyallon E, Varoquaux G. Why do tree-based models still outperform deep learning on typical tabular data? In: Advances in Neural Information Processing Systems 35 (NeurIPS), Datasets and Benchmarks Track; 2022.

[7] Weber M, Domeniconi G, Chen J, Weidele DKI, Bellei C, Robinson T, et al. Anti-money laundering in Bitcoin: experimenting with graph convolutional networks for financial forensics. In: KDD Workshop on Anomaly Detection in Finance; 2019. arXiv:1908.02591.

[8] Tang J, Hua F, Gao Z, Zhao P, Li J. GADBench: revisiting and benchmarking supervised graph anomaly detection. In: Advances in Neural Information Processing Systems 36 (NeurIPS), Datasets and Benchmarks Track; 2023.

[9] Kaufman S, Rosset S, Perlich C, Stitelman O. Leakage in data mining: formulation, detection, and avoidance. ACM Transactions on Knowledge Discovery from Data. 2012;6(4):15.

[10] Kapoor S, Narayanan A. Leakage and the reproducibility crisis in machine-learning-based science. Patterns. 2023;4(9):100804.

[11] Bauder RA, Khoshgoftaar TM. Medicare fraud detection using machine learning methods. In: Proceedings of the 16th IEEE International Conference on Machine Learning and Applications (ICMLA); 2017. p. 858–865.

[12] Herland M, Khoshgoftaar TM, Bauder RA. Big data fraud detection using multiple Medicare data sources. Journal of Big Data. 2018;5:29.

[13] Johnson JM, Khoshgoftaar TM. Medicare fraud detection using neural networks. Journal of Big Data. 2019;6:63.

[14] Thornton D, Mueller RM, Schoutsen P, van Hillegersberg J. Predicting healthcare fraud in Medicaid: a multidimensional data model and analysis techniques for fraud detection. Procedia Technology. 2013;9:1252–1264.

[15] Akoglu L, Tong H, Koutra D. Graph based anomaly detection and description: a survey. Data Mining and Knowledge Discovery. 2015;29(3):626–688.

[16] Pourhabibi T, Ong KL, Kam BH, Boo YL. Fraud detection: a systematic literature review of graph-based anomaly detection approaches. Decision Support Systems. 2020;133:113303. https://doi.org/10.1016/j.dss.2020.113303.

[17] Akoglu L, McGlohon M, Faloutsos C. OddBall: spotting anomalies in weighted graphs. In: Advances in Knowledge Discovery and Data Mining (PAKDD). Lecture Notes in Computer Science, vol. 6119. Springer; 2010. p. 410–421.

[18] Beutel A, Xu W, Guruswami V, Palow C, Faloutsos C. CopyCatch: stopping group attacks by spotting lockstep behavior in social networks. In: Proceedings

of the 22nd International Conference on World Wide Web (WWW); 2013. p. 119–130.

[19] Hooi B, Song HA, Beutel A, Shah N, Shin K, Faloutsos C. FRAUDAR: bounding graph fraud in the face of camouflage. In: Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining; 2016. p. 895–904.

[20] Wang J, Wen R, Wu C, Huang Y, Xiong J. FdGars: fraudster detection via graph convolutional networks in online app review system. In: Companion Proceedings of the 2019 World Wide Web Conference; 2019. p. 310–316.

[21] Zhang Y, Fan Y, Ye Y, Zhao L, Shi C. Key player identification in underground forums over attributed heterogeneous information network embedding framework. In: Proceedings of the 28th ACM International Conference on Information and Knowledge Management (CIKM); 2019. p. 549–558.

[22] Dou Y, Liu Z, Sun L, Deng Y, Peng H, Yu PS. Enhancing graph neural network-based fraud detectors against camouflaged fraudsters. In: Proceedings of the 29th ACM International Conference on Information and Knowledge Management (CIKM); 2020. p. 315–324.

[23] Liu Y, Ao X, Qin Z, Chi J, Feng J, Yang H, et al. Pick and choose: a GNN-based imbalanced learning approach for fraud detection. In: Proceedings of the Web Conference 2021 (WWW); 2021. p. 3168–3177.

[24] Tang J, Li J, Gao Z, Li J. Rethinking graph neural networks for anomaly detection. In: Proceedings of the 39th International Conference on Machine Learning (ICML). PMLR, vol. 162; 2022. p. 21076–21089.

[25] Altman E, Blanuša J, von Niederäusern L, Egressy B, Anghel A, Atasu K. Realistic synthetic financial transactions for anti-money laundering models. In: Advances in Neural Information Processing Systems 36 (NeurIPS), Datasets and Benchmarks Track; 2023.

[26] Jesus S, Pombal J, Alves D, Cruz A, Saleiro P, Ribeiro R, et al. Turning the tables: biased, imbalanced, dynamic tabular datasets for ML evaluation. In: Advances in Neural Information Processing Systems 35 (NeurIPS), Datasets and Benchmarks Track; 2022.

[27] Shchur O, Mumme M, Bojchevski A, Günnemann S. Pitfalls of graph neural network evaluation. In: NeurIPS Workshop on Relational Representation Learning; 2018. arXiv:1811.05868.

[28] Xu K, Hu W, Leskovec J, Jegelka S. How powerful are graph neural networks? In: International Conference on Learning Representations (ICLR); 2019.

[29] Ke G, Meng Q, Finley T, Wang T, Chen W, Ma W, et al. LightGBM: a highly efficient gradient boosting decision tree. In: Advances in Neural Information Processing Systems 30 (NeurIPS); 2017. p. 3146–3154.

[30] Hamilton WL, Ying R, Leskovec J. Inductive representation learning on large graphs. In: Advances in Neural Information Processing Systems 30 (NeurIPS); 2017. p. 1024–1034.

[31] Pedregosa F, et al. Scikit-learn: machine learning in Python. Journal of Machine Learning Research. 2011;12:2825–2830.

[32] Fey M, Lenssen JE. Fast graph representation learning with PyTorch Geometric. In: ICLR Workshop on Representation Learning on Graphs and Manifolds; 2019. arXiv:1903.02428.

[33] Saito T, Rehmsmeier M. The precision-recall plot is more informative than the ROC plot when evaluating binary classifiers on imbalanced datasets. PLoS ONE. 2015;10(3):e0118432.

[34] Holm S. A simple sequentially rejective multiple test procedure. Scandinavian Journal of Statistics. 1979;6(2):65–70.

[35] Demšar J. Statistical comparisons of classifiers over multiple data sets. Journal of Machine Learning Research. 2006;7:1–30.

[36] Gupta RA. Healthcare Provider Fraud Detection Analysis (dataset). Kaggle; 2019. CC0 1.0. Accessed 20 October 2026. Available from: https://www.kaggle.com/datasets/rohitrox/healthcare-provider-fraud-detection-analysis.

[37] Rossi E, Chamberlain B, Frasca F, Eynard D, Monti F, Bronstein M. Temporal graph networks for deep learning on dynamic graphs. arXiv:2006.10637; 2020.

[38] Walonoski J, Kramer M, Nichols J, Quina A, Moesel C, Hall D, et al. Synthea: an approach, method, and software mechanism for generating synthetic patients and the synthetic electronic health care record. Journal of the American Medical Informatics Association. 2018;25(3):230–238.

[39] Kapoor S, Cantrell EM, Peng K, et al. REFORMS: consensus-based recommendations for machine-learning-based science. Science Advances. 2024;10(18):eadk3452.

[40] Gebru T, Morgenstern J, Vecchione B, Vaughan JW, Wallach H, Daumé III H, et al. Datasheets for datasets. Communications of the ACM. 2021;64(12):86–92.